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Unveiling Attractor Cycles in Large Language Models: A Dynamical Systems View of Successive Paraphrasing

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Abstract

Dynamical systems theory provides a framework for analyzing iterative processes and evolution over time. Within such systems, repetitive transformations can lead to stable configurations, known as attractors, including fixed points and limit cycles. Applying this perspective to large language models (LLMs), which iteratively map input text to output text, provides a principled approach to characterizing long-term behaviors. Successive paraphrasing serves as a compelling testbed for exploring such dynamics, as paraphrases re-express the same underlying meaning with linguistic variation. Although LLMs are expected to explore a diverse set of paraphrases in the text space, our study reveals that successive paraphrasing converges to stable periodic states, such as 2-period attractor cycles, limiting linguistic diversity. This phenomenon is attributed to the self-reinforcing nature of LLMs, as they iteratively favour and amplify certain textual forms over others. This pattern persists with increasing generation randomness or alternating prompts and LLMs. These findings underscore inherent constraints in LLM generative capability, while offering a novel dynamical systems perspective for studying their expressive potential.

1 Introduction

Dynamical systems theory provides a mathematical framework for understanding how iterative processes evolve over time (Sutherland, 1974; Michel et al., 2008). In such systems, repetitive transformation can guide the state of the system toward stable configurations, known as attractors (Milnor, 1985). These attractors can manifest as fixed points, limit cycles, or more complex structures. Applying this perspective to large language models, which iteratively map input text to output text, allows us to characterize their long-term behavioral patterns in a principled manner.

T_0: *This way, the filibuster serves as a balance to prevent 51 senators from steamrolling the opposition.*

...

T_7: *In this way, the filibuster serves as a safeguard to prevent a majority of 51 senators from dominating the minority.*

T_8: *In this manner, the filibuster acts as a protection to stop a simple majority of 51 senators from overpowering the minority.*

T_9: *In this way, the filibuster serves as a safeguard to prevent a simple majority of 51 senators from dominating the minority.*

...

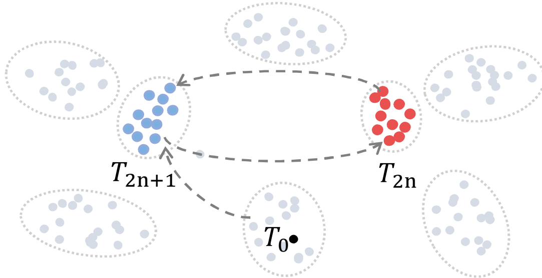

Figure 1: An illustration of successive paraphrasing using GPT-4o-mini: Here, T_0 denotes the original human-written text, while T_i indicates the i-th round of paraphrases. The nodes depicted in the lower section represent valid paraphrases for the input sentence, with distance reflecting textual variation. Successive paraphrases generated by LLMs are confined to alternating between two limited clusters, represented as blue and orange nodes.

Paraphrase generation can serve as a valuable testbed for exploring these dynamics. Paraphrases are re-expressions of the same underlying meaning, differing only in their textual or linguistic form (Bhagat and Hovy, 2013). They serve multiple purposes: improving the readability of text for language learners (Motlagh et al.; Roe and Perkins, 2022a; Kim et al., 2024), enriching datasets in low-resource scenarios (Okur et al., 2022a; Sobrevilla Cabezudo et al., 2024), and enhancing stylistic variation (Krishna et al., 2020). With recent advances in LLMs (Touvron et al., 2023; Taori et al., 2023; Brown et al., 2020; OpenAI, 2023),

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machine-generated paraphrases can rival or surpass human quality, exhibiting remarkable generalization across diverse domains and text lengths.

While producing a single paraphrase demonstrates an LLM's ability to exploit its prior knowledge to create textual variety while preserving semantic equivalance, **successive paraphrasing** pushes this capacity further. Instead of generating just one re-expression, the model recursively paraphrases its own output over multiple rounds (Sadasivan et al., 2023; Tripto et al., 2023). Intuitively, this iterative process is expected to explore an expansive linguistic landscape, generating a rich tapestry of forms. Each subsequent paraphrase, based on previously transformed text, could theoretically diverge into increasingly varied structures—similar to depth-first exploration of the paraphrase search space in contrast to breadth-first approaches like beam search (Holtzman et al., 2020a; Huang et al., 2023; Meister et al., 2023).

In practice, however, we find that this expected variety does not materialize. Instead of diverging across a vast combinatorial space, the LLM's successive paraphrasing converges onto a limited set of recurring solutions, as depicted in Figure 1. When studied through the lens of dynamical systems, these recurring solutions resemble *a stable attractor cycle—a low-order periodic orbit in the space of possible paraphrases* (Milnor, 1985). Rather than continuously discovering new linguistic configurations, the model settles into a pattern where the paraphrased outputs repeat with a fixed period. This phenomenon is subtle: it does not always manifest as explicit repetition but rather as a recurring rotation among a small set of structurally similar forms. Such periodic attractors challenge the intuition that longer or more complex texts should accomodate a broad array of distinct paraphrases. Instead, the LLM gravitates toward a small closed orbit, revealing inherent limitations in its expressive variability.

Specifically, to investigate this attractor-like behavior, we compile a diverse collection of human-written texts (Li et al., 2023) and prompt a range of both open-source and commercial LLMs to perform 15 rounds of successive paraphrasing. Using normalized Levenshtein distance to quantify textual variation, we consistently observe a **2-period cycle**: each new paraphrase resembles the one generated two steps prior. This periodicity proves robust, remaining consistent across multiple models, text lengths, and prompts. We further analyze model perplexity and generation diversity as successive paraphrasing unfolds. The results indicate that, rather than wandering freely in the paraphrase space, LLMs grow increasingly confident in a narrow set of solutions, effectively collapsing onto these attractors. Modifying generation hyperparameters or introducing perturbations, such as alternating prompts and models, only subtly disrupts these obstinate attractor cycles. Moreover, this tendency to settle into attractor cycles extends beyond paraphrasing. Any invertible task, i.e., one that allows reconstructions of previous inputs, shows similar behavior, suggesting that such cycles are a general characteristic of LLM iterative behavior.

Finally, we propose a straightforward method to disrupt attractor cycles while maintaining semantic fidelity. By intervening in the iterative process, we can reintroduce meaningful variation and prevent the model from settling into stable yet constrained periodic orbits. In summary, we propose to leverage successive paraphrasing to reveal that LLM outputs, when treated as a dynamical system, tend to converge onto stable attractor cycles rather than exploring open-ended linguistic variety. Understanding these attractors and identifying strategies to escape them is key to unlocking the full expressive potential of LLMs. We will release our data and code after the anonymous period.

2 Successive Paraphrasing as System Function

In this section, we briefly introduce the theoretical framework of dynamical systems and applies it to understand the iterative process of successive paraphrasing. By viewing paraphrase generation as the repeated application of a transformation (the LLM's paraphrasing function), we connect observed phenomena, e.g., periodicity and convergence, to well-studied concepts in systems theory.

2.1 Systems Theory Foundations

Systems theory provides a broad mathematical and conceptual framework for analyzing how complex processes evolve over time (Sutherland, 1974). The core idea is modeling the state of a system and its evolution through deterministic or stochastic rules. In continuous or discrete time, systems can exhibit distinct behaviors, ranging from stable equilibria to oscillatory dynamics or even chaotic patterns.

A **dynamical system** is commonly defined as a set of states and a rule describing how those states vary under iteration. When a transformation repeatedly maps an initial state to a new state, one of several outcomes often emerges: *Fixed Points*: States that remain unchanged under the transformation, representing equilibrium; *Limit Cycles*: Closed loops of states that recur periodically, representing sustained oscillations; *More Complex Attractors*: Patterns to which the system's trajectories converge, including chaotic attractors.

These attractors shape the long-term behavior of the system. If an initial state lies within the basin of attraction of a limit cycle, for example, the system will converge to that cycle regardless of small perturbations. Identifying such attractors offers valuable insights into the stability and variability of the system's evolution.

2.2 Framing Successive Paraphrasing as a Dynamical System

Successive paraphrasing involves iteratively generating variations of a given text while maintaining semantic equivalence, where each iteration builds upon the previous output. We propose viewing successive paraphrasing as a discrete dynamical system. Let $\mathcal{T}$ be the space of all possible texts. Consider a large language model that defines a paraphrasing function: $P : T \to T$, where $P(T)$ outputs a paraphrase of the input text T. Given an initial text $T_0 \in \mathcal{T}$, successive paraphrasing generates the sequence $\{T_n\}_{n=0}^{\infty}$ recursively by:

$$T_{n+1} = P(T_n), \quad n = 0, 1, 2, \ldots \tag{1}$$

The set $\mathcal{P}(T)$ denotes the complete text space for valid paraphrases of T, which is assumed as a finite space. In theory, the space of potential paraphrases $\mathcal{P}(T)$ can be vast, especially as text length grows. Each new iteration can potentially explore fresh textual variations, e.g., new syntactic structures, vocabulary choices, and stylistic nuances, while maintaining semantic equivalence. From a systems perspective, if the mapping P is capable of diversifying output states, one might expect the generated text sequence to spread broadly through the space $\mathcal{P}(T)$, never stuck in repetitive patterns, resembling a system without stable attractors. In contrast, if the LLM's internal biases lead to favouring certain textual forms, the sequence may enter a basin of attraction and converge onto a stable set of states. In other words, rather than exhibiting limitless variety, the system might find itself drawn to limit cycles, i.e., periodic attractors in the paraphrase space.

3 Experiment Setup

To systematically investigate this pattern, we first build dedicated testbeds and evaluation criteria.

Source Data Collection. We consider English and Chinese paraphrasing in this work. For English paraphrase generation, we collect human-written source documents by sampling instances from the MAGE dataset (Li et al., 2023). Specifically, we uniformly collect 1,000 sentences and 30 paragraphs from each domain in the dataset. This results in a total of 1,000 sentences and 300 paragraphs for subsequent paraphrasing. For Chinese, we source 200 sentences from WMT 2019 (Barrault et al., 2019) and 200 sentences from Wikipedia (Foundation). Detailed data statistics is presented in Appendix A. The main experiments (Section 4) utilize sentence-level paraphrasing datasets, while analytic experiments employ paragraph-level datasets to demonstrate the generality of our findings (Section 5).

Paraphrase Generation. For English paraphrasing, we utilize Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Meta-Llama-3-8B-Instruct (Touvron et al., 2023), Meta-Llama-3-70B-Instruct (Touvron et al., 2023), Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct (Yang et al., 2024), GPT-4o-mini and GPT-4o (OpenAI et al., 2024). For Chinese, we use Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct, and GPT-4o-mini for paraphrase generation. By default, we set the temperature to 0.6 and p to 0.9 during the decoding process. We sample 10 different paraphrases at each step by setting the number of search beams to 10 and sequentially rephrasing each sample for 15 rounds. We select the candidate with the highest probability for the next paraphrasing iteration.

Evaluation Metrics. We use the normalized Levenshtein edit distance function d to quantify the textual differences between two paraphrases. To provide a more intuitive of the attractor cycle, we propose a metric termed 2-periodicity degree to quantify and study the cyclic pattern in successive paraphrasing. The 2-periodicity degree τ is defined as $\tau = 1 - \frac{1}{M-2}\sum_{i=3}^{M} d(T_i, T_{i-2})$, which captures the average textual similarity between the current paraphrase and that from two steps prior. M denotes the total number of paraphrasing iterations. A higher τ indicates stronger periodicity, i.e., similar